Anthony Kuang

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Education

University of California, Berkeley **Graduation:** Dec 2027
B.A. in Computer Science **Major GPA:** 3.85

Coursework: Natural Language Processing† · Computer Vision† (Exam rank: 3/54) · Reinforcement Learning† · LLM Seminar† · Stochastic Processes† · Machine Learning (Exam rank: 5/404) · Robotics · Optimization · Data Structures † Graduate level

Experience

Berkeley Artificial Intelligence Research, Darrell Group *Research Assistant* Apr 2026 – Present

- Investigating whether mid-training a unified multimodal model (BAGEL) on paired graphics code and rendered images improves reasoning capabilities across modalities, targeting second-author conference submission.

Deft Robotics *Robotics Intern, San Francisco* May - Aug 2026

- Trained Diffusion Policy and Action Chunking Transformer (ACT) models on ∼150 teleoperated demonstrations for precise automotive parts insertion into millimeter-scale slots, deployed and evaluated on a UR7e robot arm.
- Supervised rollouts, logged edge cases, and collected teleoperation data for 6 clients across ∼30 applications.

Stripe *SWE Intern, San Francisco* May - Aug 2025

- Contributed to internal Node.js package for analyzing taxpayer data from a REST API, migrated client portals across 10+ internal teams to said package; integrated with internal services using TypeScript, Jest, React, Express.js.
- Developed Python heuristics to prioritize test execution in Jenkins CI pipelines for the largest Ruby codebases (30M+ LOC, 2M+ tests). Built visualizations and dashboards to evaluate test-ordering strategies.

Code Ninjas *Code Sensei, Livermore CA* Feb 2025 – Present

- Work 18 hrs/week as a Lua and Python instructor, guiding young students through coding projects and curriculum.

CS 70: Discrete Math and Probability *Undergraduate Course Staff 1* Jun - Aug 2024

- Taught ∼300 students math concepts, host office hours, grade student work, and answer questions on an online forum.

Publications

Probing Layerwise Sensitivity for LLM Security Analysis Collaborators: Ahmed Ismail, Ayo Akinkugbe, Sean O'Brien, Kevin Zhu
Cognitive Behavioral Modeling with Activation Steering

- Developed an interpretability framework that steers LLM internal activations, tracing how cognitive and security biases (e.g., jailbreaks, misinformation) emerge across layers and identifying tipping points where interventions flip behavior.
- PLSLSA accepted to NeurIPS Lock-LLM Workshop, CBMAS accepted to NeurIPS CogInterp Workshop in 2025.

Hybrid Algorithms Consistently Beat Offline RLHF Methods Collaborators: Richik Pal, Eddie Cui

- Implemented LLM RL algorithms (DPO, IPO, AOT, GRPO, DrGRPO, GSPO) on Qwen2.5-1.5B-Instruct and designed a weighted hybrid AOT-DPO objective with reward model reranking, reaching a 0.865 ± 0.015 win rate over base model.

WaveletFlow: Spatially-Localized Frequency Mixing for Flow Matching in PDE Turbulence Collaborator: Shuai Meng

- Improved FourierFlow, a leading generative model for simulating fluid flows by incorporating wavelets in the architecture to better represent high frequency features (shears and vortices), reducing prediction error by 8.4%.

Visual–Temporal RAG for Lecture Question Answering Collaborators: Micah Mok, Meenakshi Mittal

- Developed a multimodal RAG pipeline for lecture video QA that aligns speech transcripts, slide OCR, and vision-language frame descriptions, improving accuracy from 20.4% to 82.0% on a curated 156-question test set.

Projects

Viso *RAG, FAISS, Manim, LaTeX*

- AI education platform generating visual lectures with 160M token RAG corpus, won most technical project in AdventureX (China's largest hackathon) and placed top 20/150 in amber.ac competition for pre-seed/seed startups in Shenzhen.

AI Entrepreneurs at Berkeley Founding member of student-run AI incubator with $40M portfolio valuation.